

```

1  /* SECTION 1: Import*/
2  FILENAME REFFILE '/home/u64366633/sasuser.v94/MBA_Master.xlsx';
3
4  PROC IMPORT DATAFILE=REFFILE
5      DBMS=XLSX
6      OUT=WORK.WIDE
7      REPLACE;
8      GETNAMES=YES;
9  RUN;
10
11 proc datasets lib=work nolist;
12     modify wide;
13     rename Custid=custid Pr1=pr1 Pr2=pr2 Pr3=pr3 Pr4=pr4
14         Pr5=pr5 Pr6=pr6 Pr7=pr7 Pr8=pr8;
15 quit;
16
17 proc transpose data=wide out=long (rename=(col1=PRODUCTS) drop=_label_);
18     by custid;
19     var pr1-pr8;
20 run;
21
22 title "SECTION 1 -";
23 PROC CONTENTS DATA=WORK.WIDE; RUN;
24 title;

```

SECTION 1 -

The CONTENTS Procedure

Data Set Name	WORK.WIDE	Observations	500
Member Type	DATA	Variables	9
Engine	V9	Indexes	0
Created	04/29/2026 21:06:37	Observation Length	192
Last Modified	04/29/2026 21:06:37	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information

Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	682
Obs in First Data Page	500
Number of Data Set Repairs	0
Filename	/saswork/SAS_workA7200001552C_0daws02-usw2-2.oda.sas.com/SAS_work88260001552C_0daws02-usw2-2.oda.sas.com/wide.sas7bdat
Release Created	9.0401MB
Host Created	Linux
Inode Number	67116164
Access Permission	rw-r--r--
Owner Name	u64366633
File Size	256KB
File Size (bytes)	262144

Owner Name	u64366633
File Size	256KB
File Size (bytes)	262144

Alphabetic List of Variables and Attributes

#	Variable	Type	Len	Format	Informat	Label
1	custid	Num	8	BEST.		Custid
2	pr1	Char	24	\$24.	\$24.	Pr1
3	pr2	Char	24	\$24.	\$24.	Pr2
4	pr3	Char	24	\$24.	\$24.	Pr3
5	pr4	Char	24	\$24.	\$24.	Pr4
6	pr5	Char	24	\$24.	\$24.	Pr5
7	pr6	Char	24	\$24.	\$24.	Pr6
8	pr7	Char	24	\$24.	\$24.	Pr7
9	pr8	Char	14	\$14.	\$14.	Pr8

```

24  title;
25  /* SECTION 2: Product Customer Count + Macro Variables */
26  proc sql;
27      create table product_customer_count as
28      select products, count(distinct(custid)) as ANALYSIS_UNIT_FREQ
29      from long
30      where products ne ""
31      group by 1;
32  quit;
33
34  proc sql noprint;
35      select count(distinct(products)) into: product_count from long where products ne "";
36      select count(distinct(custid)) into: population from long;
37
38      select distinct products
39      into :product_1 - :product_%trim(&product_count)
40      from product_customer_count
41      where products ne "";
42
43      select distinct analysis_unit_freq
44      into :analysis_unit_freq_1 - :analysis_unit_freq_%trim(&product_count)
45      from product_customer_count
46      where products ne ""
47      order by products;
48  quit;
49
50  title "SECTION 2 - product_customer_count";
51  proc print data=product_customer_count; run;
52  title;

```

SECTION 2 - product_customer_count

Obs	PRODUCTS	ANALYSIS_UNIT_FREQ
1	BTL_Mortgage	157
2	Business_Current_Account	222
3	Credit_Card	457
4	Currency_Services	35
5	Insurance	356
6	Locker	262
7	Personal_Current_Account	288
8	Personal_Loans	218
9	Premium_Current_Account	132
10	Residential_Mortgage	215
11	Savings_Account	339
12	Trading_Account	96

```

53  title;
54  /* SECTION 3: Frequency Co-Occur - Figure 7.4 */
55  %macro product_tot;
56  %do i = 1 %to &product_count;
57      Proc sql;
58          create table product_tot_&i as
59          select products, count(distinct(custid)) as FREQ_CO_OCCUR
60          from
61          (select custid, products
62           from long
63           where custid in (select custid from long where products eq "&product_&i"))
64          where products ne ""
65          group by 1
66          order by 1;
67      quit;
68  %end;
69  %mend;
70  %product_tot;
71
72  %put product is &product_1;
73
74  title "SECTION 3 - product_tot_1, LHS = &product_1 (Figure 7.4)";
75  proc print data=product_tot_1; run;
76  title;

```

SECTION 3 - product_tot_1, LHS = BTL_Mortgage (Figure 7.4)

Obs	PRODUCTS	FREQ_CO_OCCUR
1	BTL_Mortgage	157
2	Business_Current_Account	94
3	Credit_Card	147
4	Currency_Services	9
5	Insurance	97
6	Locker	84
7	Personal_Current_Account	60
8	Personal_Loans	53
9	Premium_Current_Account	46
10	Residential_Mortgage	43
11	Savings_Account	110
12	Trading_Account	17

```

76 /* SECTION 4: Metrics + Relevant 2-Way Rules - Figure 7.5 */
77 %macro metrics;
78 %do i = 1 %to &product_count;
79   proc sql;
80     create table metric_&i as
81     select distinct "&iproduct_&i" as LHS, a.PRODUCTS as RHS, a.ANALYSIS_UNIT_FREQ,
82           b.freq_co_occur,
83           (b.FREQ_CO_OCCUR/((a.analysis_unit_freq_&i)*100) as CONFIDENCE format 5.2,
84           (b.FREQ_CO_OCCUR/((a.population)*100) as SUPPORT format 5.2,
85           (a.ANALYSIS_UNIT_FREQ/((a.population)*100) as EXPECTED_CONFIDENCE format 5.2,
86           calculated CONFIDENCE/calculated EXPECTED_CONFIDENCE as LIFT format 5.2,
87           (FREQ_CO_OCCUR*(calculated LIFT-1)**2)*((calculated SUPPORT/100)*(calculated CONFIDENCE/100))
88           /
89           (((calculated CONFIDENCE/100 - calculated SUPPORT/100)*(calculated LIFT - calculated CONFIDENCE/100)) as CHISQ format 5.2,
90           1 - Probchi(calculated CHISQ,1) as P format 5.4
91     from product_customer_count as a left join product_tot_&i as b
92     on a.products=b.products
93     order by calculated P;
94   quit;
95 %end;
96 %mend;
97 %metrics;
98
99 proc sql noprint;
100   select memname into :datasets separated by ' '
101   from dictionary.tables
102   where libname = "WORK" and memname like "METRIC%";
103 quit;
104
105 data summary;
106   format LHS $50.;
107   set &datasets;
108   if LHS ne RHS;
109 run;
110
111 proc sql noprint;
112   select memname into :datasets separated by ' '
113   from dictionary.tables
114   where libname = "WORK" and memname like "METRIC%";
115 quit;
116
117 data summary;
118   format LHS $50.;
119   set &datasets;
120   if LHS ne RHS;
121 run;
122
123 data relevant (drop=analysis_unit_freq);
124   set summary;
125   if freq_co_occur ge 50;
126   if P le 0.05;
127   if confidence ge 60;
128   if lift gt 1;
129 run;
130
131 proc sort data=relevant;
132   by descending lift;
133 run;
134
135 title "SECTION 4 - RELEVANT 2-way rules (Figure 7.5)";
136 proc print data=relevant; run;
137 title;

```

SECTION 4 - RELEVANT 2-way rules (Figure 7.5)

Obs	LHS	RHS	FREQ_CO_OCCUR	CONFIDENCE	SUPPORT	EXPECTED_CONFIDENCE	LIFT	CHISQ	P
1	Premium_Current_Account	Personal_Loans	93	70.45	18.60	43.60	1.62	9.78	.0018
2	Premium_Current_Account	Business_Current_Account	93	70.45	18.60	44.40	1.59	9.17	.0025
3	Premium_Current_Account	Locker	106	80.30	21.20	52.40	1.53	11.87	.0006
4	Personal_Loans	Personal_Current_Account	158	72.48	31.60	57.60	1.26	11.07	.0009
5	Savings_Account	Personal_Current_Account	244	71.98	48.80	57.60	1.25	43.48	.0000
6	Personal_Current_Account	Savings_Account	244	84.72	48.80	67.80	1.25	43.48	.0000
7	Business_Current_Account	Locker	145	65.32	29.00	52.40	1.25	7.74	.0054
8	Savings_Account	Locker	208	61.36	41.60	52.40	1.17	14.09	.0002
9	Locker	Savings_Account	208	79.39	41.60	67.80	1.17	14.09	.0002
10	Savings_Account	Credit_Card	321	94.69	64.20	91.40	1.04	9.31	.0023
11	Credit_Card	Savings_Account	321	70.24	64.20	67.80	1.04	9.31	.0023

```

123 title "SECTION 4 - RELEVANT 2-way rules (Figure 7.5)";
124 proc print data=relevant; run;
125 title;
126 /* SECTION 5: Condense - Figure 7.6 */
127 data condense;
128   set relevant;
129   if lhs not in ("Personal_Current_Account",
130                "Savings_Account",
131                "Credit_Card",
132                "Locker",
133                "Personal_Loans");
134 run;
135
136 title "SECTION 5 - CONDENSE (Figure 7.6)";
137 proc print data=condense; run;
138 title;

```

SECTION 5 - CONDENSE (Figure 7.6)

Obs	LHS	RHS	FREQ_CO_OCCUR	CONFIDENCE	SUPPORT	EXPECTED_CONFIDENCE	LIFT	CHISQ	P
1	Premium_Current_Account	Personal_Loans	93	70.45	18.60	43.60	1.62	9.78	.0018
2	Premium_Current_Account	Business_Current_Account	93	70.45	18.60	44.40	1.59	9.17	.0025
3	Premium_Current_Account	Locker	106	80.30	21.20	52.40	1.53	11.87	.0006
4	Business_Current_Account	Locker	145	65.32	29.00	52.40	1.25	7.74	.0054

```

139 /* SECTION 6: 3-Way Metrics - Figure 7.7 */
140
141 /* Build matrix of all valid 3-product sequences */
142 data one;
143   input id $ product1 $50.;
144   datalines;
145 1 BTL_Mortgage|
146 2 Business_Current_Account|
147 3 Credit_Card|
148 4 Currency_Services|
149 5 Insurance|
150 6 Locker|
151 7 Personal_Current_Account|
152 8 Personal_Loans|
153 9 Premium_Current_Account|
154 10 Residential_Mortgage|
155 11 Savings_Account|
156 12 Trading_Account|
157 ;
158
159 data two;
160   input id $ product2 $50.;
161   datalines;
162 1 BTL_Mortgage|
163 2 Business_Current_Account|
164 3 Credit_Card|
165 4 Currency_Services|
166 5 Insurance|
167 6 Locker|
168 7 Personal_Current_Account|
169 8 Personal_Loans|
170 9 Premium_Current_Account|

```

```

171 10 Residential_Mortgage|
172 11 Savings_Account|
173 12 Trading_Account|
174 ;
175
176 data three;
177     input id $ product3 $50.;
178     datalines;
179 1 BTL_Mortgage
180 2 Business_Current_Account
181 3 Credit_Card
182 4 Currency_Services
183 5 Insurance
184 6 Locker
185 7 Personal_Current_Account
186 8 Personal_Loans
187 9 Premium_Current_Account
188 10 Residential_Mortgage
189 11 Savings_Account
190 12 Trading_Account
191 ;
192 run;
193
194 data stage1;
195     set one;
196     do i = 1 to n;
197         set two point=i nobs=n;
198         output;
199     end;
200 run;
201
202 data stage2;
203     *** *****
204
205
210 data matrix (drop = lhand1 lhand2);
211     set stage2 (drop = id);
212     if product1 ne product2;
213     if product1 ne product3;
214     if product2 ne product3;
215     combo = compress(product1 || product2 || product3);
216     lhand1 = scan(combo, 1);
217     lhand2 = scan(combo, 2);
218     lhand = compress(lhand1 || "|" || lhand2);
219 run;
220
221 /* Pattern finding */
222 proc sql noprint;
223     select count(distinct(combo)) into :combo_count from matrix;
224     select distinct combo into :combo_1 - :combo_&trim(&combo_count) from matrix;
225     select count(distinct(lhand)) into :lhand_count from matrix;
226     select distinct lhand into :lhand_1 - :lhand_&trim(&lhand_count) from matrix;
227 quit;
228
229 %macro combo_find;
230 %do i = 1 %to &combo_count;
231     proc sql;
232         create table combos_main_&i as
233         select a.*,
234             compress(pr1||"|"||pr2||"|"||pr3||"|"||pr4||"|"||pr5||"|"||pr6||"|"||pr7||"|"||pr8) as combo,
235             "&combo_&i" as pattern_found,
236             case when calculated combo contains "&combo_&i" then 1 else 0 end as combo_count
237         from wide as a;
238
239         create table combos_sum_&i as
240         select pattern_found, sum(combo_count) as freq_co_occur
241         from combos_main_&i

```

SECTION 6 - METRIC_THREE_WAY (Figure 7.7)

Obs	LHS	RHS	ANALYSIS_UNIT_FREQ	freq_co_occur	CONFIDENCE	SUPPORT	EXPECTED_CONFIDENCE	LIFT	CHISQ	P
1	Personal_Current_Account[Savings_Account	Credit_Card	121	56	46.28	11.20		91.40	0.51	46.30 .0000
2	Insurance[Business_Current_Account	Credit_Card	57	11	19.30	2.20		91.40	0.21	9.36 .0022
3	Residential_Mortgage[Insurance	Credit_Card	71	47	66.20	9.40		91.40	0.72	6.29 .0122
4	Locker[Savings_Account	Credit_Card	55	31	56.36	6.20		91.40	0.62	5.98 .0144
5	Insurance[Trading_Account	Credit_Card	42	12	28.57	2.40		91.40	0.31	5.53 .0187
6	BTL_Mortgage[Insurance	Credit_Card	46	22	47.83	4.40		91.40	0.52	5.38 .0203
7	Business_Current_Account[Locker	Savings_Account	91	13	14.29	2.60		67.80	0.21	3.79 .0514
8	Personal_Loans[Insurance	Credit_Card	45	4	8.89	0.80		91.40	0.10	3.43 .0642
9	Premium_Current_Account[Business_Current_Account	Locker	40	40	100.0	8.00		52.40	1.91	3.16 .0755
10	Savings_Account[Residential_Mortgage	Credit_Card	39	22	56.41	4.40		91.40	0.62	2.90 .0886
11	Insurance[Credit_Card	Savings_Account	74	24	32.43	4.80		67.80	0.48	2.39 .1222
12	Credit_Card[Personal_Loans	Savings_Account	71	23	32.39	4.60		67.80	0.48	2.19 .1393
13	Personal_Current_Account[Credit_Card	Insurance	99	52	52.53	10.40		71.20	0.74	2.18 .1395
14	Personal_Loans[Premium_Current_Account	Business_Current_Account	51	40	78.43	8.00		44.40	1.77	2.13 .1443
15	Credit_Card[Insurance	Personal_Current_Account	95	28	29.47	5.60		57.60	0.51	2.13 .1447
16	Credit_Card[Insurance	Locker	95	17	17.89	3.40		52.40	0.34	1.90 .1677
17	Personal_Current_Account[Savings_Account	Personal_Loans	121	15	12.40	3.00		43.60	0.28	1.90 .1685
18	Locker[Residential_Mortgage	Credit_Card	26	5	19.23	1.00		91.40	0.21	1.82 .1776
19	Insurance[Personal_Current_Account	Savings_Account	55	51	92.73	10.20		67.80	1.37	1.79 .1804
20	Personal_Current_Account[Savings_Account	Residential_Mortgage	121	22	18.18	4.40		43.00	0.42	1.77 .1840
21	Premium_Current_Account[Locker	Credit_Card	37	2	5.41	0.40		91.40	0.06	1.50 .2201
22	Business_Current_Account[Locker	Insurance	91	3	3.30	0.60		71.20	0.05	1.50 .2205
23	Trading_Account[Business_Current_Account	Credit_Card	23	5	21.74	1.00		91.40	0.24	1.49 .2225
24	Business_Current_Account[Currency_Services	Trading_Account	13	13	100.0	2.60		19.20	5.21	1.46 .2269
25	Residential_Mortgage[BTL_Mortgage	Credit_Card	32	20	62.50	4.00		91.40	0.68	1.45 .2280
26	Personal_Current_Account[Credit_Card	Locker	99	6	6.06	1.20		52.40	0.12	1.28 .2558
27	Personal_Current_Account[Credit_Card	Residential_Mortgage	99	11	11.11	2.20		43.00	0.26	1.13 .2885
28	Credit_Card[Locker	Savings_Account	56	22	39.29	4.40		67.80	0.58	1.03 .3094
29	Credit_Card[Insurance	Residential_Mortgage	95	18	18.95	3.60		43.00	0.44	1.00 .3181
30	Residential_Mortgage[Insurance	Personal_Current_Account	71	6	8.45	1.20		57.60	0.15	0.98 .3217
31	Savings_Account[Credit_Card	Personal_Loans	117	5	4.27	1.00		43.60	0.10	0.96 .3270
32	Residential_Mortgage[Insurance	Locker	71	9	12.68	1.80		52.40	0.24	0.94 .3317
33	Savings_Account[Credit_Card	BTL_Mortgage	117	50	42.74	10.00		31.40	1.36	0.91 .3398
34	Business_Current_Account[Insurance	Credit_Card	39	1	2.56	0.20		91.40	0.03	0.85 .3567
35	Residential_Mortgage[Credit_Card	Insurance	37	10	27.03	2.00		71.20	0.38	0.76 .3832
36	Personal_Current_Account[Trading_Account	Business_Current_Account	17	17	100.0	3.40		44.40	2.25	0.75 .3867
37	Personal_Current_Account[Credit_Card	Personal_Loans	99	27	27.27	5.40		43.60	0.63	0.72 .3953
38	Insurance[Residential_Mortgage	Business_Current_Account	31	24	77.42	4.80		44.40	1.74	0.70 .4026
39	Business_Current_Account[Insurance	Savings_Account	39	8	20.51	1.60		67.80	0.30	0.69 .4051
40	BTL_Mortgage[Personal_Loans	Premium_Current_Account	24	16	66.67	3.20		26.40	2.53	0.67 .4119
41	Locker[Insurance	Business_Current_Account	42	29	69.05	5.80		44.40	1.56	0.65 .4185
42	Residential_Mortgage[Business_Current_Account	Insurance	35	6	17.14	1.20		71.20	0.24	0.64 .4224
43	Insurance[Locker	Savings_Account	44	4	9.09	0.80		67.80	0.13	0.61 .4350
44	Insurance[Business_Current_Account	Locker	57	8	14.04	1.60		52.40	0.27	0.61 .4358
45	Personal_Loans[Insurance	Savings_Account	45	3	6.67	0.60		67.80	0.10	0.51 .4760
46	Insurance[Credit_Card	Locker	74	27	36.49	5.40		52.40	0.70	0.48 .4902
47	Residential_Mortgage[BTL_Mortgage	Insurance	32	12	37.50	2.40		71.20	0.53	0.45 .5002
48	Credit_Card[Personal_Loans	Residential_Mortgage	71	5	7.04	1.00		43.00	0.16	0.44 .5088
49	Locker[Personal_Current_Account	Currency_Services	4	4	100.0	0.80		7.00	14.29	0.43 .5127
50	Credit_Card[Locker	Insurance	56	32	57.14	6.40		71.20	0.80	0.39 .5329
51	Personal_Current_Account[Savings_Account	BTL_Mortgage	121	3	2.48	0.60		31.40	0.08	0.37 .5420
52	Premium_Current_Account[Locker	Savings_Account	37	16	43.24	3.20		67.80	0.64	0.35 .5523
53	Credit_Card[Insurance	Business_Current_Account	95	2	2.11	0.40		44.40	0.05	0.34 .5599
54	Locker[Insurance	Savings_Account	42	2	4.76	0.40		67.80	0.07	0.33 .5634
55	BTL_Mortgage[Credit_Card	Personal_Current_Account	36	8	22.22	1.60		57.60	0.39	0.32 .5728
84	Residential_Mortgage[Insurance	Business_Current_Account	71	1	1.41	0.20		44.40	0.03	0.12 .7248
85	BTL_Mortgage[Savings_Account	Personal_Current_Account	9	9	100.0	1.80		57.60	1.74	0.12 .7275
86	Credit_Card[BTL_Mortgage	Locker	52	1	1.92	0.20		52.40	0.04	0.12 .7306
87	Personal_Loans[Premium_Current_Account	Locker	51	1	1.96	0.20		52.40	0.04	0.12 .7336
88	Savings_Account[Residential_Mortgage	Personal_Current_Account	39	17	43.59	3.40		57.60	0.76	0.12 .7339
89	Locker[Insurance	Residential_Mortgage	42	2	4.76	0.40		43.00	0.11	0.11 .7408
90	Credit_Card[Insurance	Trading_Account	95	11	11.58	2.20		19.20	0.60	0.10 .7559
91	BTL_Mortgage[Business_Current_Account	Locker	30	10	33.33	2.00		52.40	0.64	0.09 .7604
92	Savings_Account[Personal_Loans	Insurance	15	6	40.00	1.20		71.20	0.56	0.09 .7666
93	Savings_Account[Premium_Current_Account	Locker	25	17	68.00	3.40		52.40	1.30	0.09 .7676
94	Residential_Mortgage[Personal_Current_Account	Credit_Card	21	21	100.0	4.20		91.40	1.09	0.09 .7685
95	Insurance[Business_Current_Account	Personal_Loans	57	29	50.88	5.80		43.60	1.17	0.08 .7768
96	Residential_Mortgage[Insurance	Trading_Account	71	5	7.04	1.00		19.20	0.37	0.08 .7789
97	Savings_Account[Personal_Current_Account	Credit_Card	43	41	95.35	8.20		91.40	1.04	0.08 .7821
98	Business_Current_Account[Credit_Card	Locker	21	5	23.81	1.00		52.40	0.45	0.07 .7887
99	Personal_Loans[Insurance	Business_Current_Account	45	1	2.22	0.20		44.40	0.05	0.07 .7895
100	Premium_Current_Account[Personal_Loans	Insurance	9	9	100.0	1.80		71.20	1.40	0.07 .7962
101	Personal_Current_Account[Premium_Current_Account	Credit_Card	5	1	20.00	0.20		91.40	0.22	0.07 .7980
102	Credit_Card[Personal_Current_Account	Savings_Account	8	8	100.0	1.60		67.80	1.47	0.06 .8037
103	Credit_Card[Savings_Account	Premium_Current_Account	42	15	35.71	3.00		26.40	1.35	0.06 .8043
104	Currency_Services[Trading_Account	Credit_Card	16	13	81.25	2.60		91.40	0.89	0.06 .8124
105	Locker[Residential_Mortgage	Insurance	26	21	80.77	4.20		71.20	1.13	0.05 .8206
106	Savings_Account[Premium_Current_Account	Personal_Loans	25	6	24.00	1.20		43.60	0.55	0.05 .8242
107	Savings_Account[Insurance	Locker	23	15	65.22	3.00		52.40	1.24	0.05 .8272
108	Savings_Account[Personal_Loans	Premium_Current_Account	15	7	46.67	1.40		26.40	1.77	0.05 .8306
109	Credit_Card[Residential_Mortgage	Insurance	11	10	90.91	2.00		71.20	1.28	0.04 .8365
110	Savings_Account[BTL_Mortgage	Business_Current_Account	4	4	100.0	0.80		44.40	2.25	0.04 .8407
111	Locker[BTL_Mortgage	Insurance	7	7	100.0	1.40		71.20	1.40	0.04 .8411
112	Credit_Card[Locker	Premium_Current_Account	56	1	1.79	0.20		26.40	0.07	0.04 .8428
113	Business_Current_Account[BTL_Mortgage	Insurance	15	8	53.33	1.60		71.20	0.75	0.04 .8444
114	Credit_Card[BTL_Mortgage	Business_Current_Account	52	26	50.00	5.20		44.40	1.13	0.04 .8448

```

324  /* SECTION 7: Relaxed 3-Way Rules - Figure 7.8 */
325  data relevant_three_way;
326      set metric_three_way;
327      if confidence ge 40;
328      if lift gt 1;
329      if freq_co_occur ge 25;
330  run;
331
332  title "SECTION 7 - RELEVANT 3-way relaxed (Figure 7.8)";
333  proc print data=relevant_three_way (drop=chisq p); run;
334  title;

```

home/u64366633/sasuser.v94/Lab8.sas

SECTION 7 - RELEVANT 3-way relaxed (Figure 7.8)

Obs	LHS	RHS	ANALYSIS_UNIT_FREQ	freq_co_occur	CONFIDENCE	SUPPORT	EXPECTED_CONFIDENCE	LIFT
1	Premium_Current_Account Business_Current_Account	Locker	40	40	100.0	8.00	52.40	1.91
2	Personal_Loans Premium_Current_Account	Business_Current_Account	51	40	78.43	8.00	44.40	1.77
3	Insurance Personal_Current_Account	Savings_Account	55	51	92.73	10.20	67.80	1.37
4	Savings_Account Credit_Card	BTL_Mortgage	117	50	42.74	10.00	31.40	1.36
5	Locker Insurance	Business_Current_Account	42	29	69.05	5.80	44.40	1.56
6	Insurance Business_Current_Account	Personal_Loans	57	29	50.88	5.80	43.60	1.17
7	Savings_Account Personal_Current_Account	Credit_Card	43	41	95.35	8.20	91.40	1.04